1 From Attention to FlashAttention

Both standard Attention and FlashAttention share the same input: Q, K, V. Both of them perform softmax normalization to transform their raw attention scores to probabilities in between 0 and 1, and both of them perform max normalization to account for the risks of floating-point overflow in practice. Now, suppose Q is a $A \times B$, K is $C \times B$, and V is $C \times D$, and the final output matrix O, which is the matrix of weighted attention scores, is $A \times D$. Then:

1.1 Standard Attention

Standard Attention computes raw scores S, a $A \times C$ matrix, by $S = \frac{QK^T}{\sqrt{B}}$. S is then normalized with softmax, which scales the arbitrarily large scores of S into probabilities between 0 and 1, producing a matrix of weighted attention scores, P. P shares the exact same dimensions as S, and the probabilities in each row of P add up to 1. Formally, $P = softmax_{row}(S)$, where $P_{ij} = \frac{e^{S_{ij}}}{\sum_k e^{S_{ik}}}$. To account for the risks of overflow as raw scores can be arbitrarily large, P is then normalized with the max raw score m_i: $P_{ij} = \frac{e^{S_{ij}-m_i}}{\sum_k e^{S_{ik}-m_i}}$. Lastly, the output matrix O from the standard attention is computed as $O = PV \in \mathbb{R}^{A \times D}$.

So, with input Q, K, V, standard Attention computes output matrix $O = PV$ with instantiations of two $\mathbb{R}^{\mathbb{A} \times \mathbb{B}}$ matrices as intermediary states, S and P, where:

$S = \frac{QK^T}{\sqrt{d}}$, $P = softmax_{row}(S)$ with $P_{ij} = \frac{e^{S_{ij}-m_i}}{\sum_k e^{S_{ik}-m_i}}$ in practice.

1.2 FlashAttention

FlashAttention optimizes Attention through NOT instantiating the two $\mathbb{R}^{A \times D}$ matrices, S and P, as intermediary states. Formally, FlashAttention does online computation for each output row in the output matrix O through $O_i = \frac{\sum_j e^{S_{ij}-m_i} V_j}{\sum_j e^{S_{ij}-m_i}}$. This can be rewritten equivalently as $O_i = \frac{\sum_j e^{S_{ij}-m_i} V_i}{\sum_j e^{S_{ij}-m_i}} = \frac{N_i}{l_i}$ with two intermediary values, $l_i N_i$, for each output row O_i, where:

$l_i = \sum_j e^{S_{ij}-m_i}$, $N_i = \sum_j e^{S_{ij}-m_i} V_i, m_i = max_{1 \le j \le C} S_{ij}$

Hence $O_i = \frac{N_i}{l_i}$, and N_i, l_i are normalized with m_i, all of which are updated at each new iteration of computation. In other words, m_i is the running max raw score, l_i is the running denominator in the online fraction that computes for one output row, and N_i is the running numerator in the same online fraction.

Since FlashAttention is an online computation, we also define it recursively as:

For $m_i = max_{1 \le j \le C} S_{ij}$, define $m_i^{(0)} = -\infty$ and $m_i^{(j)} = max(m_i^{(j-1)}, S_{ij})$

For $l_i = \sum_j e^{S_{ij}-m_i}$, define, $l_i^{(0)} = 0$ and $l_i^{(j)} = e^{m_i^{(j-1)} - m_i^{(j)}} l_i^{(j-1)} + e^{S_{ij}-m_i^{(j)}}$

For $N_i = \sum_j e^{S_{ij}-m_i} V_j$, define $N_i^{(0)} = 0_D$ and $N_i^{(j)} = e^{m_i^{(j-1)}-m_i^{(j)}} N_i^{(j-1)} + e^{S_{ij}-m_i^{(j)}} V_j$

Then, after C iterations, $O_i = \frac{N_i^{(C)}}{l_i^{(C)}}$

Note, the recursive definitions for l_i (N_i as well in the same line of idea) is derived from the following:

$l_i^{(j)} = \sum_{k=1}^{j} e^{S_{ik}-m_i^{(j)}}$

$= \sum_{k=1}^{j-1} e^{S_{ik}-m_i^{(j)}} + e^{S_{ik}-m_i^{(j)}} \quad (note: e^{S_{ik}-m_i^{(j)}} = e^{(S_{ik}-m_i^{(j-1)})} e^{(m_i^{(j-1)}-m_i^{(j)})})$

$= \sum_{k=1}^{j-1} e^{S_{ik}-m_i^{(j-1)}} e^{m_i^{(j-1)}-m_i^{(j)}} + e^{S_{ik}-m_i^{(j)}}$

$= e^{m_i^{(j-1)}-m_i^{(j)}} \sum_{k=1}^{j-1} e^{S_{ik}-m_i^{(j-1)}} + e^{S_{ik}-m_i^{(j)}}$

$= e^{m_i^{(j-1)}-m_i^{(j)}} l_i^{j-1} + e^{S_{ik}-m_i^{(j)}}$

1.3 Algebraic Transformation from Attention to FlashAttention

We want to transform $O = PV = softmax_{row}(\frac{QK^T}{\sqrt{B}})V$ to $O = \begin{bmatrix} O_1 \\ O_2 \\ ... \\ O_A \end{bmatrix}$, where $O_i = \frac{\sum_j e^{S_{ij}-m_i} V_j}{\sum_j e^{S_{ij}-m_i}}$ for $1 \leq i \leq A$

Observe that the standard Attention compputes its output matrix O by row. Then, the full algebraic transformation from the standard Attention to FlashAttention, represented in recursively defined form, via rewriting becomes:

$O_i = \sum_{j=1}^{C} P_{ij} V_j$

$= \sum_{j=1}^{C} \frac{e^{S_{ij}-m_i}}{\sum_{k=1}^{C} e^{S_{ik}-m_i}} V_j$

$= \frac{\sum_{j=1}^{C} e^{S_{ij}-m_i} V_j}{\sum_{k=1}^{C} e^{S_{ik}-m_i}}$

$= \frac{N_i}{l_i}$

$= \frac{N_i^{(C)}}{l_i^{(C)}}$

Where $S_{ij} = \frac{1}{\sqrt{B}} \sum_{b=1}^{B} Q_{ib} K_{jb}$,

$m_i^{(0)} = -\infty$ and $m_i^{(j)} = max(m_i^{(j-1)}, S_{ij})$

$l_i^{(0)} = 0$ and $l_i^{(j)} = e^{m_i^{(j-1)}-m_i^{(j)}} l_i^{(j-1)} + e^{S_{ij}-m_i^{(j)}}$

$N_i^{(0)} = 0_D$ and $N_i^{(j)} = e^{m_i^{(j-1)}-m_i^{(j)}} N_i^{(j-1)} + e^{S_{ij}-m_i^{(j)}} V_j$

Then:

$$O = \begin{bmatrix} \frac{N_1^{(C)}}{l_1^{(C)}} \\ \frac{N_2^{(C)}}{l_2^{(C)}} \\ \dots \\ \frac{N_A^{(C)}}{l_A^{(C)}} \end{bmatrix} \in \mathbb{R}^{A \times D}$$

2 Generic Recurrences to WIDELETREC

2.1 Generic Recurrences to lanes

Lanes: A lane is an array L of length n, and every cell is updated at once on each tick with the same rule as the recurrence:

$$F(L)[0] = b, F(L)[0] = g(L[j-1], j) \ for \ 1 \leq j < n$$

Note that cell j reads cell $j-1$ from the previous tick's array. The machine starts from zeros and applies F once per tick:

$$L_0 = [0, 0, \dots, 0], L_{\{t+1\}} = F(L_t)$$

Generic Recurrences to Lanes: $r^{(n-1)} = L_t[n-1]$